

Logistic Regression

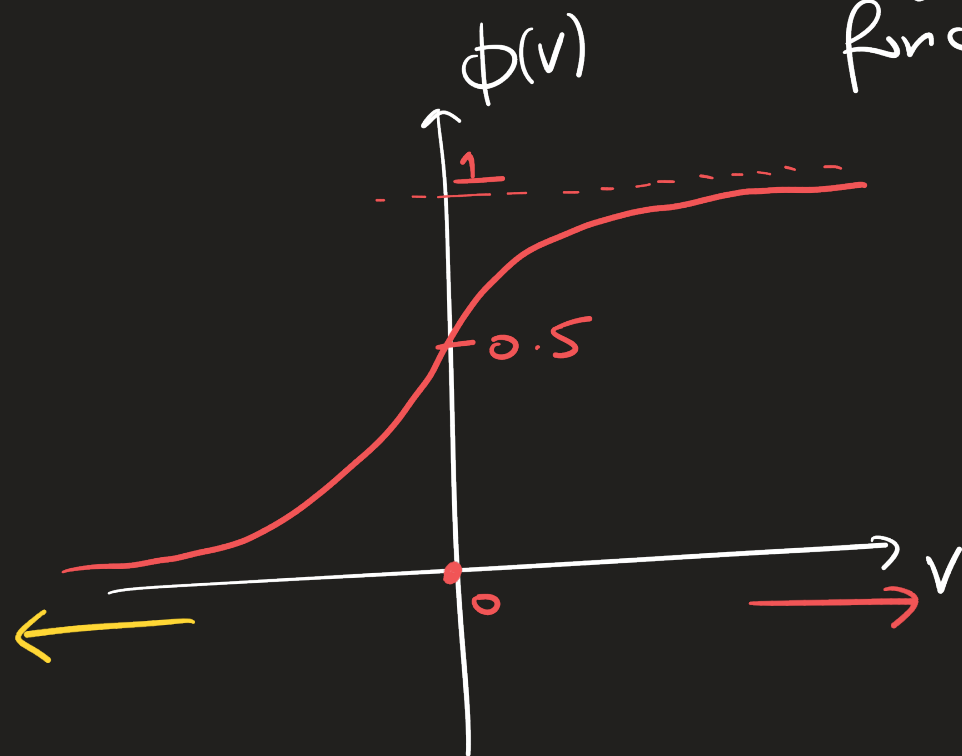
Mapper fct.: $y(x) = \phi(\omega^T x + \omega_0)$

DATA:

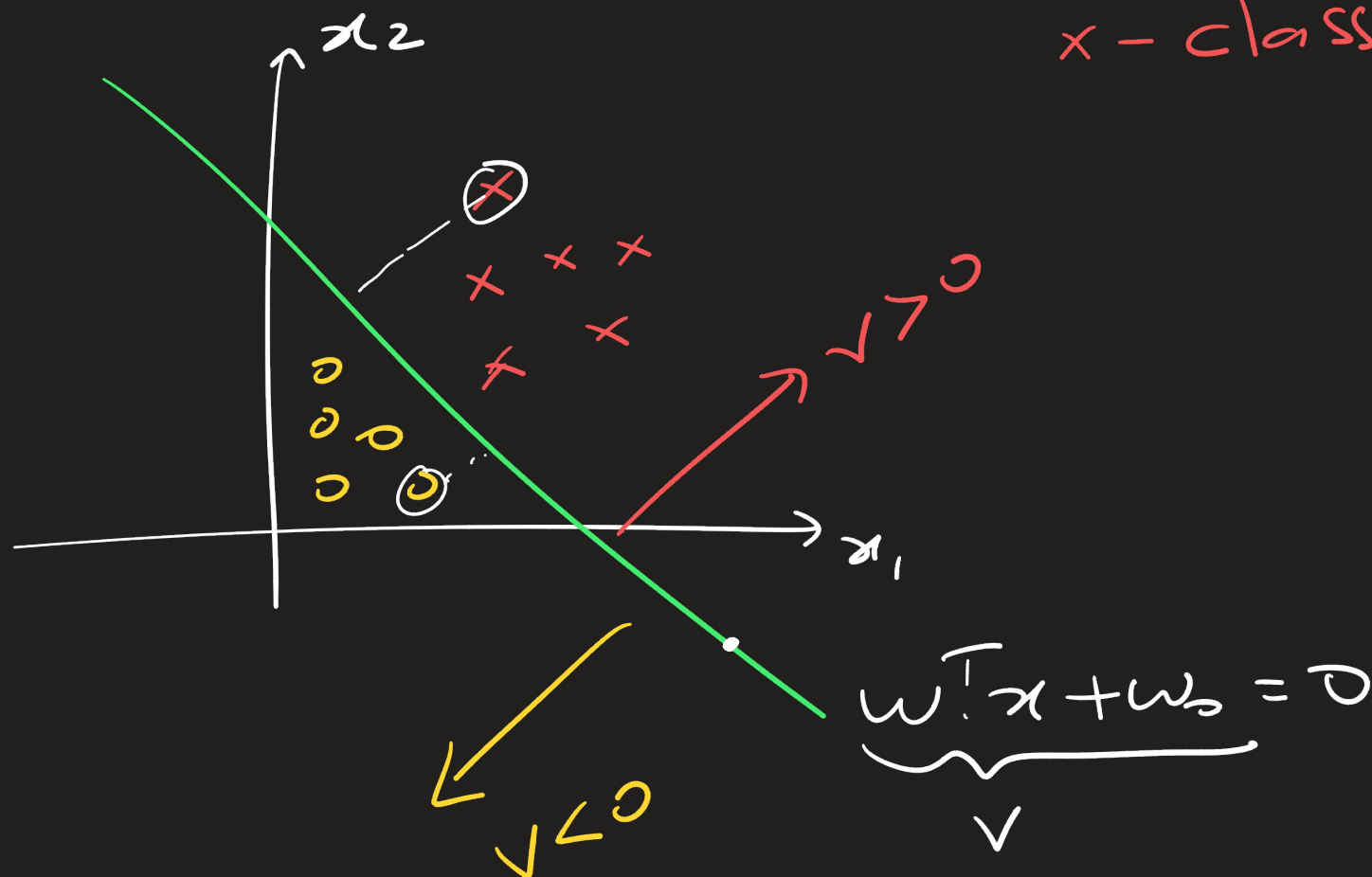
$$\{(x_i, t_i)\}_{i=1}^N$$

$$x_i \in \mathbb{R}^D, t_i \in \{0, 1\}$$

where $\phi(v) \equiv$ sigmoid or logistic function $= \frac{1}{1 + e^{-v}} \in [0, 1]$



o - class 0
x - class 1



DATASET: $\{x_1, x_2, x_3\}$

target: $\{1, 0, \underline{\underline{1}}\}$

Mapper
fct.
output

$\{0.6, 0.3, 0.4\}$

>0.5 <0.5 <0.5



1



0



0X

probability
that a
point belongs
to class 1.

1) we can define the point of operation, δ

$$\begin{cases} \phi(w^T x + w_0) < \delta, & x \in C_0 \\ \phi(w^T x + w_0) \geq \delta, & x \in C_1 \end{cases}$$

typically $\delta = 0.5$

Objective
function :

maximize probability
of assignment in the
correct class.

1) if x 's label is $t = 1$, then
we want to maximize $\phi(\omega^T x + \omega_0)$.

2) if x 's label is $t = 0$, then
we want to maximize $1 - \phi(\omega^T x + \omega_0)$

$$J(\omega, \omega_0) = \prod_{i=1}^N \phi(v_i)^{t_i} (1 - \phi(v_i))^{1-t_i}$$

i.i.d.

$t_i \in \{0, 1\} \equiv$ true label
for sample
 x_i

and

$$v_i = \omega^T x_i + \omega_0$$

$$\begin{aligned} x &= \{x_1, x_2, x_3\} \\ t &= \{1, 0, 1\} \\ y &= \{0.6, 0.3, 0.4\} \end{aligned}$$

$$J = 0.6 \times (1 - 0.3) \times 0.4$$

Applying the natural-log fct onto
 $J(w, w_0)$:

$$J(w, w_0) = \sum_{i=1}^N [t_i \cdot \ln(\phi(v_i)) + (1-t_i) \cdot \ln(1-\phi(v_i))]$$

↓
If we treat it as minimization problem,
then the new obj. fct.

$$J(w, w_0) = - \sum_{i=1}^N [t_i \cdot \ln(\phi(v_i)) + (1-t_i) \cdot \ln(1-\phi(v_i))]$$

(Binary) Cross-Entropy.

↳ Convex function →

global minimum

$$\phi(v) = \frac{1}{1 + e^{-v}}$$

We do not have an analytical
solution for w and w_0 .

So we will use an iterative
search algorithm such as

gradient descent
(1st order algorithm).

Pseudo-code:

① Iteration, $(t) = 0$.

② Initialize parameters $w^{(t)}$ and $w_0^{(t)}$.

③ Update the parameters
(online / stochastic or batch update).

$$\begin{cases} w^{(t+1)} \leftarrow w^{(t)} - \eta \nabla_w J(w^{(t)}, w_0^{(t)}) \\ w_0^{(t+1)} \leftarrow w_0^{(t)} - \eta \nabla_{w_0} J(w^{(t)}, w_0^{(t)}) \end{cases}$$

④ increment iteration: $t \leftarrow t+1$

⑤ go back to step ③ until convergence criteria is met:

a) reaching max. # epochs.

b) if $\|w^{(t+1)} - w^{(t)}\| \leq \epsilon$

$\downarrow$
e.g. 10^{-3}

$$J(\omega, \omega_0) = - \sum_{i=1}^N [t_i \cdot \ln(\phi(v_i)) + (1-t_i) \cdot \ln(1-\phi(v_i))]$$

where $\phi(v) = \frac{1}{1+e^{-v}}$, $v_i = \omega^T x_i + \omega_0$

and $\phi'(v) = \phi(v) \cdot (1-\phi(v))$

$$\nabla_{\omega} J(\omega, \omega_0) = \frac{\partial J}{\partial \omega}$$

$\underbrace{\quad}_{\text{"}/\phi(v_i) \text{ (red)"}}$
 $\underbrace{\quad}_{\text{"}\phi'(v_i) \text{ (blue)"}}$
 $\underbrace{\quad}_{\text{"}x_i \text{ (yellow)"}}$

$$= - \sum_{i=1}^N \left[t_i \cdot \frac{\partial \ln(\phi(v_i))}{\partial \phi(v_i)} \cdot \frac{\partial \phi(v_i)}{\partial v_i} \cdot \frac{\partial v_i}{\partial \omega} + \right.$$

assuming
Batch Learning

$$+ (1-t_i) \cdot \frac{\partial \ln(1-\phi(v_i))}{\partial \phi(v_i)} \cdot \frac{\partial \phi(v_i)}{\partial v_i} \cdot \frac{\partial v_i}{\partial \omega} \Bigg]$$

$\underbrace{\quad}_{= \frac{-1}{1-\phi(v_i)} \text{ (red)"}}$
 $\underbrace{\quad}_{\text{"}\phi'(v_i) \text{ (blue)"}}$
 $\underbrace{\quad}_{\text{"}x_i \text{ (yellow)"}}$